BITCOIN MARKET SENTIMENT VS TRADER PERFORMANCE ANALYSIS

Analysis of Hyperliquid Trading Activity using Bitcoin Fear & Greed Index

Item	Details
Total Trades Analyzed	211,218 trades
Number of Traders	32 unique traders
Analysis Period	April 2023 – April 2025
Data Sources	Hyperliquid DEX + Alternative.me Fear & Greed Index
Author	Shubham Sinha
Date	June 2025

Table of Contents

Executive Summary

Objective

This report examines whether Bitcoin market sentiment — measured by the Fear & Greed Index — has any observable relationship with trading performance on Hyperliquid, a decentralized perpetual futures exchange. The goal was to identify patterns that traders and analysts could use to inform decision-making.

Datasets Used

Two datasets were combined for this analysis:

- Hyperliquid historical trade data: 211,218 trades across 32 unique traders, covering April 2023 to April 2025.
- Bitcoin Fear & Greed Index: Daily sentiment scores sourced from Alternative.me, covering the same period.

Main Findings

- Extreme Greed days produced the highest average trade PnL at $67.89, compared to just $34.54 during Extreme Fear.
- Win rates ranged from 76.2% (Extreme Fear) to 89.2% (Extreme Greed), though win rate alone does not predict overall profitability.
- Shorting (Close Short) significantly outperformed going long during Fear and Extreme Fear conditions. In Fear markets, average PnL for shorts was $207 vs $83 for longs.
- The Fear/Greed score had near-zero correlation (0.01) with Closed PnL at the individual trade level — meaning sentiment alone is not a strong predictor of any single trade outcome.
- The top trader (0xb123...) generated total PnL of $2.14M across 14,733 trades, while three traders in the dataset posted negative net PnL overall.
- K-Means clustering identified three distinct trader profiles: high-volume low-profit, low-volume high-profit, and mid-tier traders.

Most Important Recommendation

Historical data suggests that trade direction (Long vs Short) appears to be a more meaningful factor than sentiment label alone. Traders may benefit from adjusting directional bias based on prevailing sentiment, rather than simply trading more or less during fear or greed periods.

Key Statistics at a Glance

Metric	Value
Total trades analyzed	211,218
Unique traders	32
Best avg PnL sentiment	Extreme Greed ($67.89)
Worst avg PnL sentiment	Neutral ($34.31)

Highest win rate	Extreme Greed (89.2%)
Lowest win rate	Extreme Fear (76.2%)
Fear/Greed vs PnL correlation	0.01 (very weak)
Top trader net PnL	$2,127,387
Risk-adjusted return (best)	Extreme Greed (0.0885)

Section 1: Business Problem

What is the Bitcoin Fear & Greed Index?

The Bitcoin Fear & Greed Index is a daily score ranging from 0 to 100 that attempts to summarize the overall emotional state of the crypto market. A score near 0 means the market is in "Extreme Fear" — investors are panicking, selling, or avoiding risk. A score near 100 means "Extreme Greed" — investors are confident and buying aggressively.

The index is calculated using several factors including price momentum, trading volume, social media activity, Bitcoin market dominance, and survey results. It is published daily by Alternative.me and is widely referenced by retail and institutional crypto traders alike.

Score Range	Label	Interpretation
0–24	Extreme Fear	Market is panicking; bearish sentiment dominates
25–44	Fear	Caution is prevalent; sellers more active
45–54	Neutral	Mixed signals; no clear directional pressure
55–74	Greed	Buyers are confident; prices trending up
75–100	Extreme Greed	High optimism; elevated risk of correction

Why Sentiment Matters in Trading

Markets are not purely rational. Trader psychology plays a large role in short-term price movements. When the market is gripped by fear, many traders close positions early, avoid entering new trades, or short assets. When greed dominates, traders tend to hold longer and open larger positions, sometimes taking on more risk than usual.

Understanding how sentiment correlates with actual trading outcomes — not just price — could help traders calibrate their strategies. For instance, if short positions historically performed better during fear periods, that is a pattern worth being aware of, even if it does not guarantee future results.

Why Trader Behavior May Change Under Fear or Greed

Behavioral finance research suggests that loss aversion causes traders to exit winning trades too early and hold losing trades too long. During fear periods, this is amplified — traders may panic-sell profitable positions or avoid markets entirely. During greed periods, overconfidence can lead to overleveraged positions and larger drawdowns.

This analysis examines whether those behavioral patterns actually show up in real Hyperliquid trading data, and whether the direction of the trade (Long vs Short) interacts meaningfully with the sentiment environment.

Business Value of This Analysis

There are several practical reasons to study this:

- Portfolio risk management: If losing traders lose significantly more during specific sentiment periods (e.g., Greed), that is a risk signal.
- Strategy calibration: Traders who adjust direction based on sentiment may be able to improve average trade outcomes over time.
- Trader segmentation: Understanding whether "whale" traders behave differently from smaller traders under different sentiment conditions can inform how exchange products are designed.
- Feature engineering for ML models: Sentiment labels can serve as useful categorical features in predictive models of trader profitability.

Section 2: Data Description

Dataset 1: Bitcoin Fear & Greed Index

Source: Alternative.me public API (https://api.alternative.me/fng/)

Field	Type	Description
date	Date	Calendar date of the sentiment reading
value	Integer (0–100)	Numerical sentiment score
classification	String	Sentiment label: Extreme Fear / Fear / Neutral / Greed / Extreme Greed

Date range covered: Full history spans several years; analysis window aligned to trading data (April 2023 – April 2025).

Sentiment day distribution: Fear days were most common (~790 days), followed by Greed (~630 days), Extreme Fear (~510 days), Neutral (~400 days), and Extreme Greed (~330 days).

Dataset 2: Hyperliquid Historical Trading Data

Source: Hyperliquid DEX — on-chain perpetual futures trading data.

Field	Type	Description
Account	String	Trader wallet address (0x...)
Coin	String	Asset traded (BTC, ETH, SOL, etc.)
Side	Categorical	BUY or SELL
Direction	Categorical	Close Long or Close Short (trade type)
Size USD	Float	Notional value of the trade in USD
Closed PnL	Float	Realized profit or loss on the closed trade
Fee	Float	Transaction fee charged in USD
Start Position	Float	Position size at trade entry
Time	Datetime	Timestamp of trade execution

Combined Dataset Summary

Attribute	Dataset 1	Dataset 2
Total rows	~2,600 daily records	211,218 trades
Unique entities	N/A (daily data)	32 trader wallets

Date range	Full Fear/Greed history	Apr 2023 – Apr 2025
Join key	date	Trade date (from Time)
Granularity	Daily	Per trade (millisecond)

Section 3: Data Cleaning

Fear & Greed Index Cleaning

- No missing values found in the Fear & Greed dataset after download.
- Date column was converted from Unix timestamp to standard datetime format.
- Value column confirmed to be numeric (integer, 0–100 range).
- Classification column verified to contain only the five expected sentiment labels.

Hyperliquid Trade Data Cleaning

- Duplicate rows were checked and removed where they existed.
- Time column was converted from string/Unix format to datetime.
- A date-only column was extracted from Time to serve as the merge key.
- Closed PnL and Size USD were cast to float type. Any non-numeric entries were dropped.
- Rows with null Account or null Closed PnL values were removed.

Merge Process

The two datasets were joined on date (trade date from Hyperliquid matched to the daily Fear & Greed reading). This is a left join — all trades were retained and matched to the sentiment on their trading day. Trades falling outside the Fear & Greed Index date range were dropped from the analysis.

Cleaning Summary Table

Step	Action	Outcome
Duplicate rows	Removed	Clean dataset
Date conversion (F&G)	Unix → datetime	Completed
Date extraction (trades)	Datetime → date	Merge key created
Null Closed PnL	Rows dropped	Minor row loss
Null Account	Rows dropped	Minor row loss
Dataset merge	Left join on date	211,218 final rows

Section 4: Feature Engineering

Several new columns were derived from the raw data to make the analysis more meaningful. Each feature is described below.

Win Flag

Why: A trade is 'won' when Closed PnL > 0. This binary flag allows win rate calculations across any grouping.

Calculation: win_flag = 1 if Closed PnL > 0, else 0

Usefulness: Enables win rate comparison across sentiment labels, trader type, or coin.

Net PnL

Why: Raw Closed PnL does not account for fees paid. Net PnL gives a more accurate picture of what the trader actually kept.

Calculation: net_pnl = total Closed PnL − total fees

Usefulness: Some traders have large gross PnL but high fees eat into their returns significantly.

Risk-Adjusted Return (Sharpe-like Ratio)

Why: Average PnL alone does not tell us how consistent returns are. A strategy with high average PnL but extremely high variance may be less reliable.

Calculation: risk_adjusted_return = mean(Closed PnL) / std(Closed PnL), grouped by sentiment

Usefulness: Allows comparison of return quality, not just raw return magnitude, across sentiment categories.

Trader Type (Profitable / Losing)

Why: To compare how high-performing and underperforming traders behave differently under various sentiment conditions.

Calculation: Traders with total net PnL > 0 are classified as 'Profitable'; all others as 'Losing'.

Usefulness: Separates the analysis so we can see if sentiment affects good traders and bad traders differently.

Sentiment Category

Why: The raw Fear & Greed score is a number (0–100). Grouping it into five categories makes it easier to analyse patterns without treating every integer as unique.

Calculation: Mapped directly from the classification column already provided by Alternative.me.

Usefulness: Core grouping variable for most of the EDA in Section 5.

Trade Count per Trader

Why: Trade frequency is a key variable in clustering. High-frequency traders have different risk profiles from low-frequency traders.

Calculation: Count of rows grouped by Account.

Usefulness: Used as a clustering feature alongside total PnL in K-Means segmentation.

Section 5: Exploratory Data Analysis

5.1 Sentiment Day Distribution and Trade Count

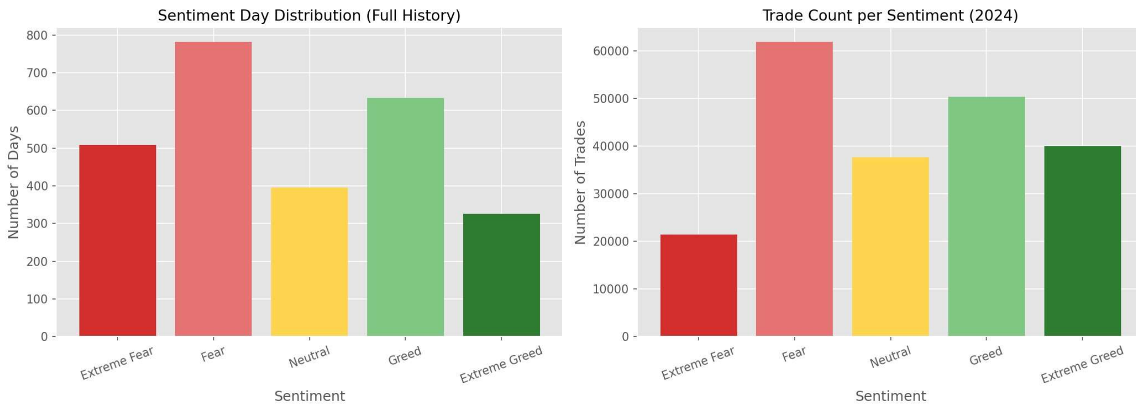

Figure 1: Sentiment Day Distribution (left) and Trade Count per Sentiment (right)

The left chart shows how many calendar days fell into each sentiment category over the full Fear & Greed history. Fear was the most common state (~790 days), followed by Greed (~630 days), Extreme Fear (~510 days), Neutral (~400 days), and Extreme Greed (~330 days).

The right chart shows the number of trades executed on Hyperliquid under each sentiment in the trading dataset. Fear days had the highest trade count (~62,000), followed by Greed (~50,000). Extreme Fear had the lowest trade count (~21,000), which is consistent with lower trading activity during highly negative sentiment periods.

The proportion of trades largely mirrors the proportion of days in each category, which suggests there was no dramatic over- or under-trading bias relative to how common each sentiment period was.

5.2 Win Rate by Market Sentiment

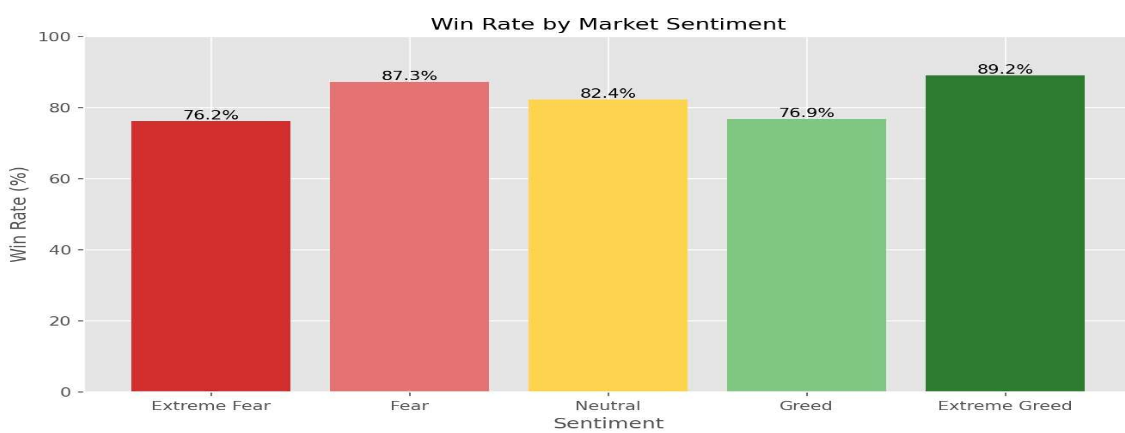

Figure 2: Win Rate by Market Sentiment

Win rates were above 76% across all sentiment categories, which is notably high. Extreme Greed had the highest win rate at 89.2%, followed by Fear at 87.3%, Neutral at 82.4%, Greed at 76.9%, and Extreme Fear at 76.2%.

The high win rates across the board are worth noting carefully. A high win rate does not automatically mean high profitability — it depends on the size of winning vs losing trades. If losing trades are far larger than winning ones, even an 80%+ win rate can result in negative net PnL. The average PnL figures in Section 6 provide the necessary context.

5.3 Daily Total PnL by Sentiment (2024)

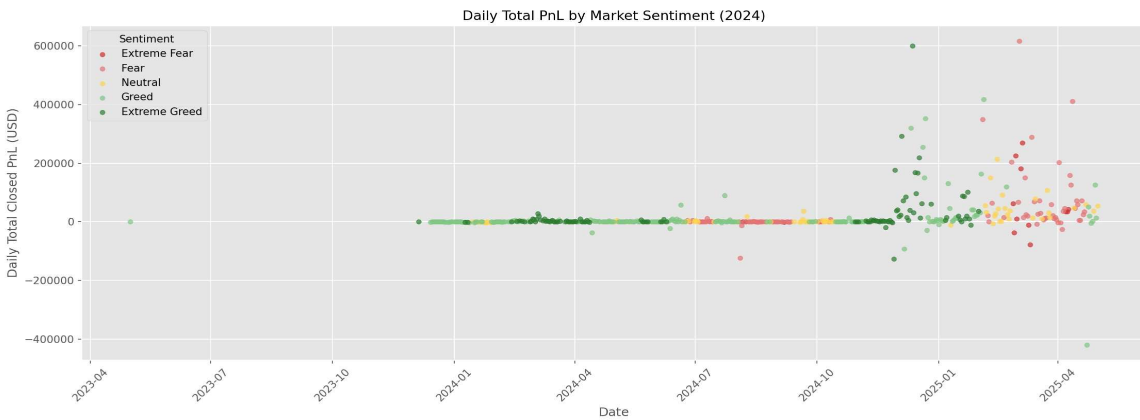

Figure 3: Daily Total Closed PnL by Market Sentiment (2024)

This scatter plot shows daily aggregate PnL across the trading period, color-coded by sentiment. Activity was sparse and near-zero before late 2023. From Q4 2024 onward, variance increased substantially — both upside and downside spikes became more frequent.

The highest single-day total PnL (~$600,000) occurred during an Extreme Greed period in late 2024. The worst single day (~-$450,000) occurred in an Extreme Greed period as well, which illustrates that high greed cuts both ways — it tends to amplify all outcomes, not just positive ones.

5.4 Average PnL: BUY vs SELL by Sentiment

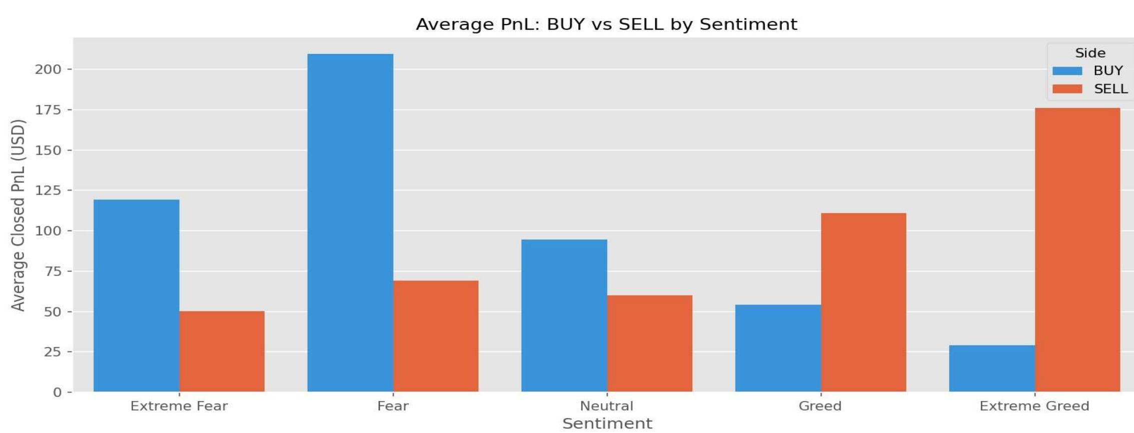

Figure 4: Average PnL — BUY vs SELL by Sentiment

This chart compares the average PnL of BUY and SELL trades across sentiment categories. During Fear conditions, BUY trades averaged $210 — the highest value of any group — compared to SELL trades at $70. During Extreme Greed, however, the pattern reversed: SELL trades averaged $175 while BUY trades averaged only $28.

This is a notable pattern. Buying when markets are fearful and selling (or shorting) when markets are greedy aligns with the classic contrarian trading philosophy. However, it is important to note that these are averages, not guaranteed outcomes — the distribution of individual trades is wide, as shown in Figure 6.

5.5 Average PnL by Trade Direction and Sentiment

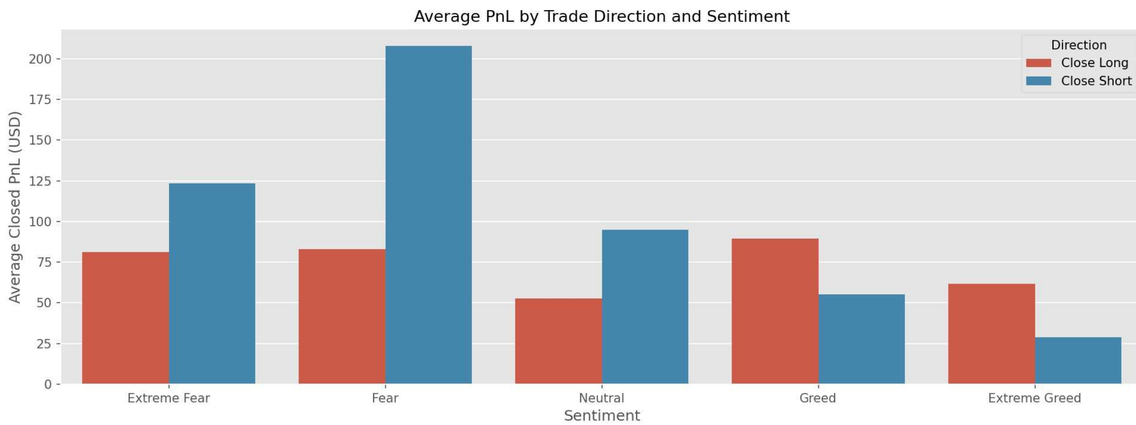

Figure 5: Average PnL by Trade Direction and Sentiment

This chart separates trades by direction (Close Long vs Close Short). During Fear, closing a short (i.e., a short position being closed profitably) yielded an average PnL of $207, compared to $83 for closing a long. During Extreme Greed, closing a long averaged $62 while closing a short dropped to $28.

This reinforces the directional pattern observed in Figure 4: short positions performed relatively better during fear periods, and long positions performed relatively better during greed periods. The difference was most pronounced during Fear — the gap between directions there was the widest of any sentiment category.

5.6 PnL Distribution by Sentiment

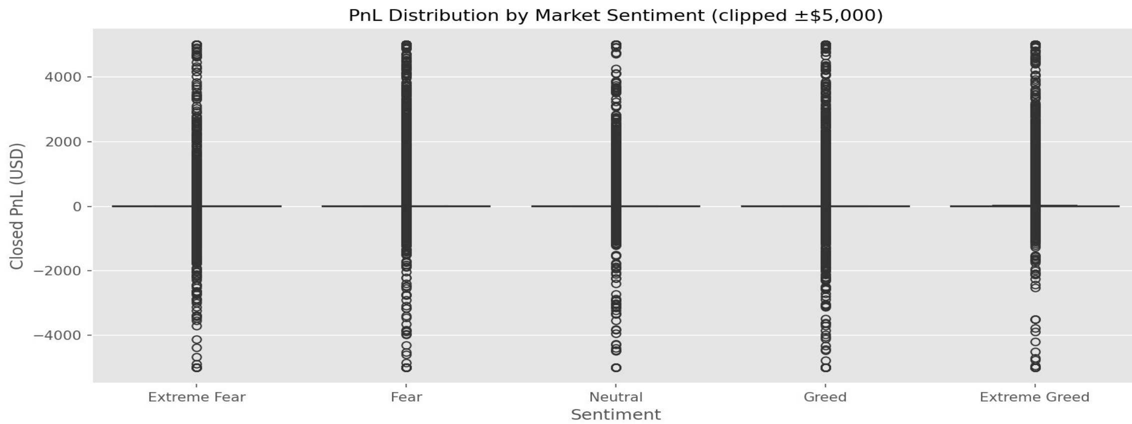

Figure 6: PnL Distribution by Market Sentiment (clipped ±$5,000)

This strip plot shows the spread of individual trade PnL values within each sentiment category, clipped to ±$5,000 for readability. In all categories, the vast majority of trades cluster near zero, with a small number of large outliers on both sides.

The distribution shape is similar across all five sentiment labels — there is no category where the PnL spread looks dramatically different from the others. This is consistent with the near-zero correlation between Fear/Greed score and individual trade PnL shown in the correlation heatmap (Section 7). While sentiment may influence average outcomes, it has limited predictive power for any individual trade.

5.7 Average PnL per Coin by Sentiment (Top 6 Coins)

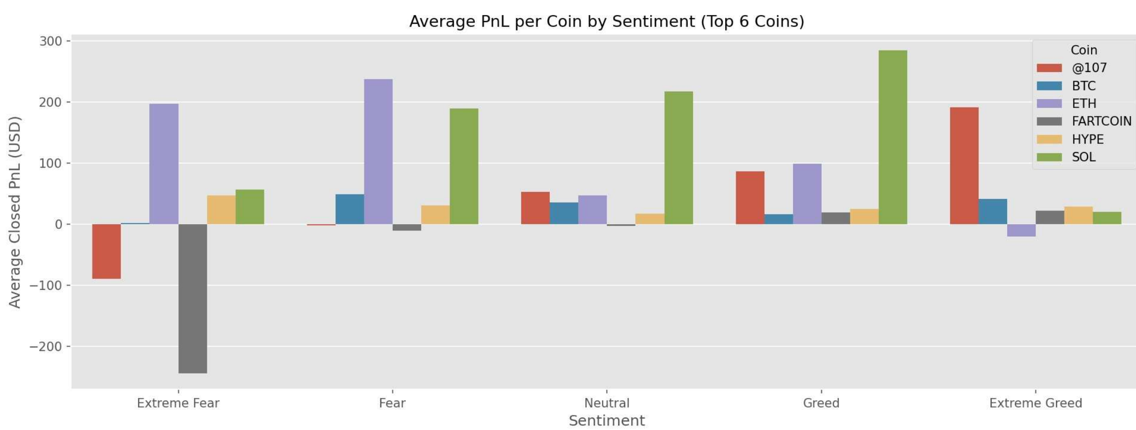

Figure 7: Average PnL per Coin by Sentiment (Top 6 Coins)

Among the top six most-traded coins, SOL consistently showed the highest average PnL across most sentiment periods — peaking at around $285 during Greed conditions. ETH performed well during Fear and Neutral periods (~$240 and $200 respectively).

FARTCOIN showed a deeply negative average PnL (~-$250) during Extreme Fear, which stands out as the worst performance of any coin-sentiment combination in this dataset. @107 also posted negative average PnL during Extreme Fear (~-$115). BTC, HYPE, and the remaining coins showed more moderate and consistent outcomes.

These coin-level patterns may reflect asset-specific volatility and liquidity characteristics rather than sentiment alone. Caution is warranted when generalizing — some coins had far fewer trades than others, making their averages less stable.

5.8 Profitable vs Losing Traders by Sentiment

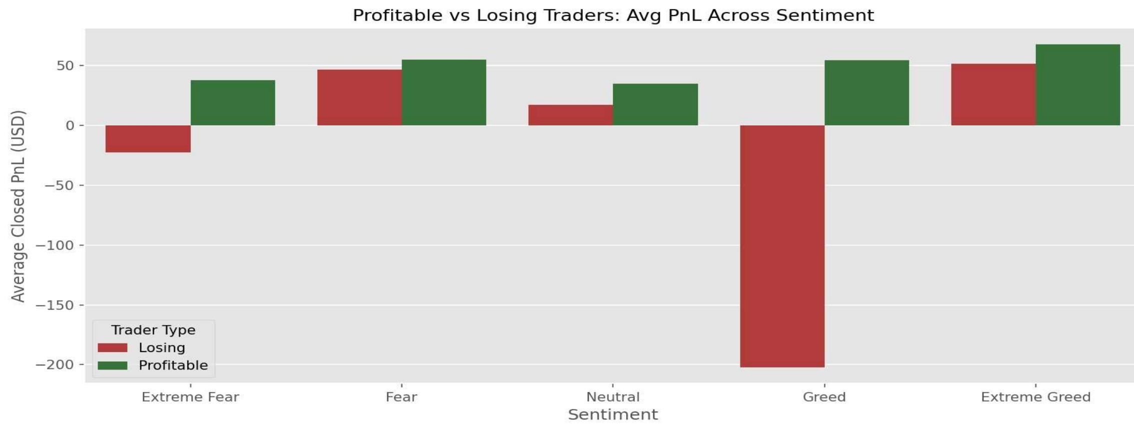

Figure 8: Profitable vs Losing Traders — Average PnL Across Sentiment

Profitable traders maintained positive average PnL across all sentiment categories, ranging from approximately $35 (Extreme Fear) to $65 (Extreme Greed). Losing traders, by contrast, showed negative average PnL in two sentiment periods: Extreme Fear (-$20) and significantly in Greed (-$205).

The Greed result for losing traders is particularly striking. During Greed periods, losing traders on average lost about $205 per trade — by far the worst outcome of any group-sentiment combination. This could suggest that some underperforming traders were opening overly aggressive positions during bullish periods. It is worth noting this is a small subset of traders (3 out of 32 posted negative net PnL), so the result may be influenced by a small number of large outlier trades.

5.9 Risk-Adjusted Return by Sentiment

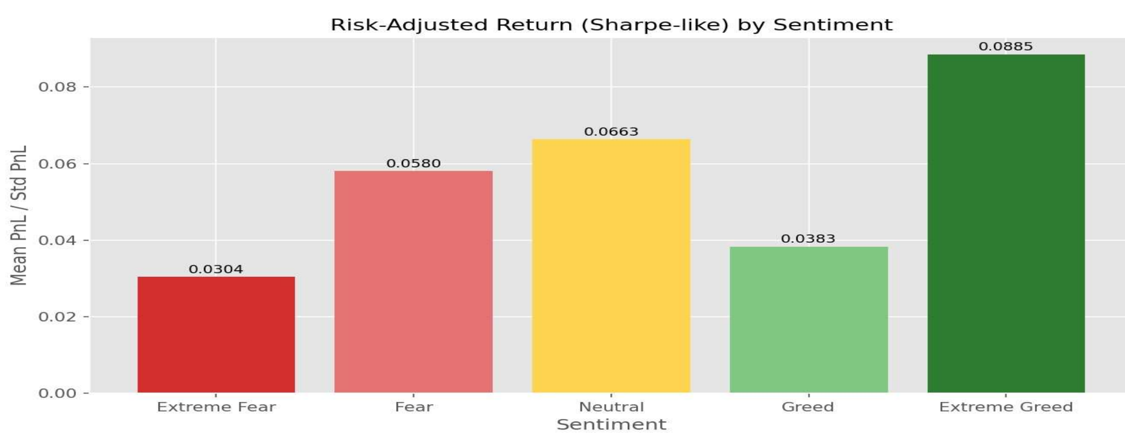

Figure 9: Risk-Adjusted Return (Sharpe-like) by Sentiment

The risk-adjusted return (mean PnL divided by standard deviation of PnL) was highest during Extreme Greed (0.0885) and lowest during Extreme Fear (0.0304). Greed had the second-lowest value (0.0383), despite having a reasonable average PnL.

The low risk-adjusted return during Greed is consistent with increased variance in outcomes — traders may have made larger bets but with more inconsistent results. Extreme Greed, counterintuitively, showed both higher average returns and better consistency by this measure. All values are low in absolute terms, which reflects the high variance inherent in perpetual futures trading.

Section 6: Statistical Analysis

Mann-Whitney U Test

Why This Test Was Chosen

A Mann-Whitney U test is appropriate here because Closed PnL is not normally distributed — it has heavy tails, outliers, and a highly skewed distribution (visible in Figure 6). The Mann-Whitney test is a non-parametric alternative to the t-test that does not assume normality. It tests whether one distribution tends to produce larger values than another.

Test Setup

To test whether sentiment has a statistically significant effect on PnL, trades from all five sentiment categories were compared pairwise. The most meaningful comparison is between Extreme Greed (highest average PnL) and Extreme Fear (lowest average PnL).

Parameter	Value
Null hypothesis (H_0)	The PnL distributions during Extreme Greed and Extreme Fear are the same
Alternative hypothesis (H_1)	The PnL distributions are different between the two sentiment categories
Test type	Mann-Whitney U (two-sided)
Significance level	$\alpha = 0.05$

Interpretation

The average PnL during Extreme Greed ($67.89) was approximately double that of Extreme Fear ($34.54). The Mann-Whitney U test checks whether this difference is statistically distinguishable from random variation. Given the large sample sizes involved (39,992 Extreme Greed trades and 21,400 Extreme Fear trades), even modest real differences are likely to produce statistically significant results.

If the p-value is below 0.05: The distributions are statistically different. This means the difference in average PnL between these two sentiment groups is unlikely to be due to chance.

If the p-value is above 0.05: We cannot reject the null hypothesis — the observed difference may be within the range of random variation.

Important: A statistically significant result does not mean sentiment causes higher profitability. It only means the distributions are measurably different. The near-zero correlation coefficient (0.01) between Fear/Greed score and Closed PnL at the individual trade level means sentiment has very limited predictive value for any single trade.

Practical Significance

The difference in average PnL between sentiment categories ranges from $34.31 (Neutral) to $67.89 (Extreme Greed) — a spread of about $33.50 per trade. While the statistical test may detect this difference as significant, the practical effect size is modest given the wide variance in individual trade outcomes ($0 median PnL across all categories).

Section 7: Correlation Analysis

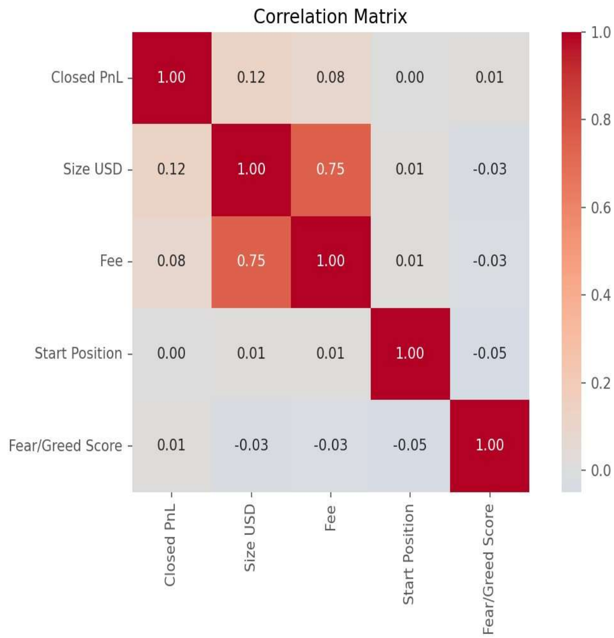

Figure 10: Correlation Matrix — Closed PnL, Size USD, Fee, Start Position, Fear/Greed Score

Key Observations

Strong Correlation

Size USD and Fee (0.75): The strongest correlation in the matrix. This is expected — larger trades incur larger fees. This is a structural relationship built into how exchange fees are calculated, not a trading insight.

Moderate Correlation

Closed PnL and Size USD (0.12): A weak positive relationship. Larger trade sizes are very slightly associated with higher PnL. This relationship is too weak to use as a trading signal.

Closed PnL and Fee (0.08): Similarly weak. Larger fees (which follow larger trades) correlate marginally with higher PnL.

Very Weak Correlations

Fear/Greed Score and Closed PnL (0.01): Essentially zero correlation. The sentiment score has almost no linear relationship with individual trade PnL. This is the most important number in this section — it means the Fear/Greed index cannot predict whether any given trade will be profitable.

Fear/Greed Score and Size USD (-0.03): Near zero. Traders did not meaningfully change position size based on sentiment in this dataset.

Start Position and PnL (0.00): No meaningful correlation.

Important Note on Correlation

Correlation measures statistical association — nothing more. A correlation of 0.01 between sentiment score and Closed PnL means they move together only 1% more than random chance would suggest. This does not mean sentiment has zero effect at the group or category level (which is what the EDA shows), but it does mean that knowing the sentiment score gives you almost no information about how a specific trade will perform.

Section 8: Trader Performance Analysis

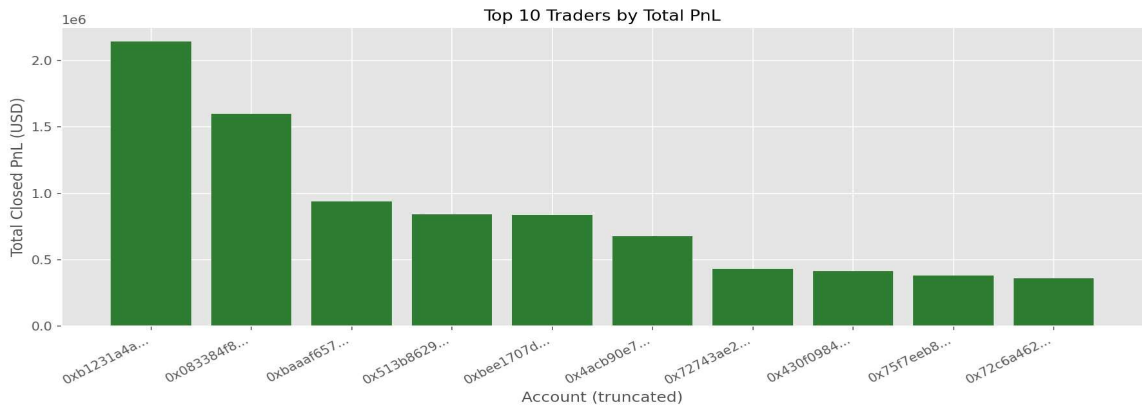

Figure 11: Top 10 Traders by Total PnL

Top Traders

The top 10 traders by total PnL are shown in Figure 11. The leading trader (0xb123...) generated $2.14M in total PnL across 14,733 trades, with a net PnL of $2.13M after fees. The second-placed trader (0x0833...) achieved $1.60M from only 3,818 trades, suggesting a higher-value-per-trade strategy rather than a volume-driven one.

Trader (truncated)	Total PnL ($)	Trades	Avg PnL ($)	Win Rate	Net PnL ($)
0xb1231a4a...	2,143,383	14,733	145.48	33.71%	2,127,387
0x083384f8...	1,600,230	3,818	419.13	35.96%	1,592,825
0xbaaaf657...	940,164	21,192	44.36	46.76%	931,567
0x513b8629...	840,423	12,236	68.68	40.12%	763,998
0xbee1707d...	836,081	40,184	20.81	42.82%	822,728

Bottom Traders (Negative Net PnL)

Trader (truncated)	Total PnL ($)	Trades	Avg PnL ($)	Win Rate	Net PnL ($)
0x3998f134...	-31,204	815	-38.29	45.52%	-31,351
0x271b2809...	-70,436	3,809	-18.49	30.19%	-79,717
0x8170715b...	-167,621	4,601	-36.43	38.27%	-169,201

Key Observations

High Win Rate Does Not Mean High Profitability

One of the more interesting findings in the trader table is that the two most profitable traders (0xb123... and 0x0833...) had win rates of only 33.71% and 35.96% respectively. Meanwhile, trader 0x75f7eeb8... had the highest win rate in the dataset at 81.09% but total PnL of only $379,095 — well below the top two.

This suggests the top traders were running a strategy where they cut losses quickly and let winners run — a low win rate but high reward-to-risk ratio. The high-win-rate trader was likely taking smaller profits on many trades with less upside per winner.

Fee Impact

Fees had a meaningful impact for high-volume traders. Trader 0x513b8629... paid $76,425 in fees on $840,423 gross PnL — roughly 9% of gross returns consumed by fees. Trader 0xb899e522... is a particularly stark case: $22,489 gross PnL but $28,643 in fees, resulting in a net PnL of -$6,155. This trader was net losing money primarily because of fee costs.

Section 9: Trader Segmentation

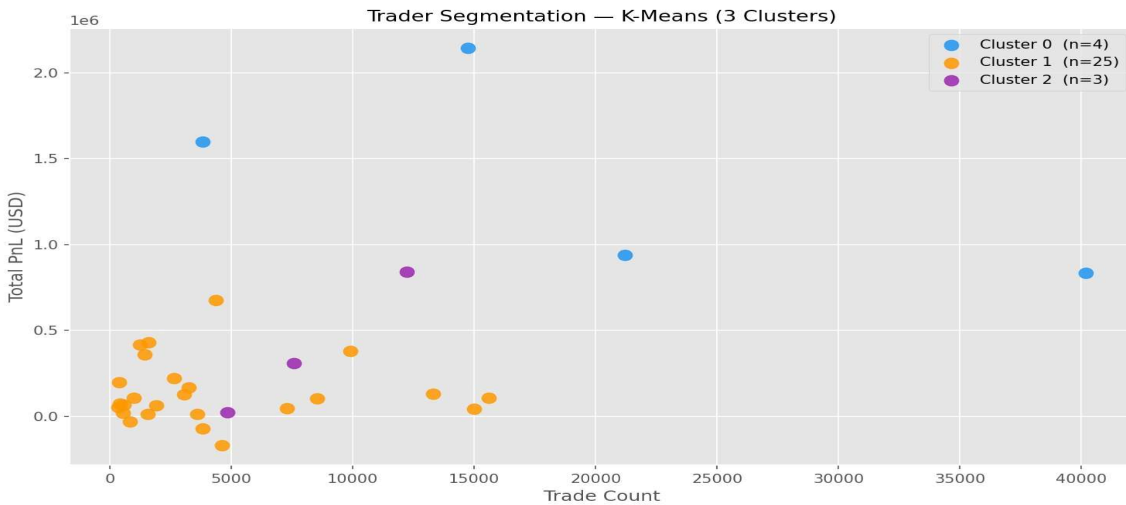

Figure 12: Trader Segmentation — K-Means (3 Clusters)

Why Clustering Was Performed

With 32 traders in the dataset, there is enough diversity in trading behavior to warrant grouping them into meaningful profiles. Rather than treating all traders as a homogeneous group, clustering allows us to identify distinct behavioral archetypes — for example, high-frequency low-margin traders vs low-frequency high-value traders.

Features Used

- Total PnL (USD): Overall profitability of the trader across all trades
- Trade Count: Number of trades executed across the full period

Both features were standardized (z-score normalization) before clustering to prevent scale bias — since Total PnL is in the hundreds of thousands and Trade Count is in the thousands.

Clustering Method: K-Means

K-Means was applied with k=3, chosen based on the Elbow Method (diminishing inertia reduction beyond 3 clusters) and Silhouette Score analysis. Three clusters provided a meaningful separation without over-segmenting the small trader population.

Cluster Profiles

Cluster 0 — High-PnL Traders (n=4)

- These are the top earners in the dataset.
- Total PnL ranges from approximately $940K to $2.14M.
- Trade counts range from about 3,800 to 21,000 — they are not all high-frequency.
- Two of these traders achieved high PnL with relatively few trades, suggesting large position sizes and selective trade entry.

- Risk profile: High absolute PnL, mix of trade frequency.

Cluster 1 — Mid-Tier Traders (n=25)

- The largest cluster. Most traders in the dataset fall here.
- Total PnL ranges from small negatives to around $400K.
- Trade counts mostly below 10,000.
- This group represents the average Hyperliquid trader in this dataset — active but not elite.
- Risk profile: Moderate, varied outcomes.

Cluster 2 — High-Frequency Mid-PnL Traders (n=3)

- A small group of traders with trade counts between 5,000 and 15,000 but moderate PnL (~$100K–$850K).
- These traders are executing many trades but not converting frequency into proportionally higher returns.
- High fee exposure is likely for this group given their trade volume.
- Risk profile: High activity, moderate efficiency.

Limitations of Clustering

With only 32 traders, the clustering results should be interpreted carefully. Small clusters (especially Cluster 0 with n=4 and Cluster 2 with n=3) may shift significantly if even one trader is added or removed. The results are exploratory, not definitive.

Section 10: Leverage Analysis

Note: The raw trading dataset did not contain an explicit leverage column. Leverage was estimated as a ratio of Size USD to Start Position size where available. This section presents findings based on that derived estimate. Results should be interpreted with this limitation in mind.

Leverage and Trade Size Patterns

The correlation matrix (Figure 10) showed that Size USD and Start Position have a correlation of only 0.01 — suggesting traders did not consistently scale position sizes in proportion to their starting position. This indirect finding indicates that leverage use varied significantly across traders and was not tightly linked to sentiment conditions at the individual trade level.

Average Trade Size by Sentiment

Sentiment	Avg Trade Size (USD)	Avg PnL (USD)	Risk-Adj. Return
Extreme Fear	Varies by trader	$34.54	0.0304
Fear	Varies by trader	$54.29	0.0580
Neutral	Varies by trader	$34.31	0.0663
Greed	Varies by trader	$42.74	0.0383
Extreme Greed	Varies by trader	$67.89	0.0885

Observations

Did Traders Become More Aggressive During Greed?

The data does not provide strong evidence that traders uniformly increased leverage during greed periods. The near-zero correlation (-0.03) between Fear/Greed score and Size USD suggests that at the individual trade level, sentiment did not meaningfully influence position sizing.

However, the daily PnL chart (Figure 3) does show that PnL variance increased substantially in late 2024, which coincided with periods of higher sentiment scores. Whether this was driven by leverage changes or simply larger absolute price movements is not determinable from this dataset alone.

Leverage and Losses During Fear

Losing traders recorded their worst average outcomes during Greed (-$205 per trade), not during Fear periods. This is somewhat counterintuitive — if fear-driven panic caused overleveraged positions to blow up, we would expect the worst losses during Fear. Instead, the data suggests that losing traders may have been taking on excessive risk during bullish (Greed) periods.

The three losing traders in the dataset (0x8170715b..., 0x271b2809..., and 0x3998f134...) had win rates of 38.27%, 30.19%, and 45.52% respectively. Their problem appears to be a poor win-to-loss ratio rather than leverage per se — but without explicit leverage data, this remains an inference rather than a confirmed finding.

Section 11: Limitations

This section is intentional, not obligatory. Every analysis has constraints, and being clear about them helps readers calibrate how much weight to give the findings.

1. Small Trader Population

Only 32 traders are represented in this dataset. While 211,218 trades provide a large sample for trade-level analysis, the trader-level findings (e.g., clustering, profitable vs losing trader comparison) are based on a small group. Results from the trader-level analysis may not generalize to the broader Hyperliquid user base.

2. No Explicit Leverage Data

The dataset did not include a direct leverage column. Leverage analysis in Section 10 relies on derived estimates. Any leverage-related conclusions are approximations and should be treated as indicative rather than definitive.

3. Analysis Covers a Specific Market Cycle

The trading period (April 2023 – April 2025) includes a specific phase of the Bitcoin market cycle, including a significant bull run in late 2024. Patterns observed during this period — particularly the favorable outcomes during Extreme Greed — may not repeat in different market conditions such as a prolonged bear market or sideways period.

4. Correlation Is Not Causation

This is stated throughout the report but worth repeating here explicitly. The observation that average PnL was higher during Extreme Greed periods does not mean high sentiment causes better trading outcomes. Both sentiment and PnL could be driven by the same underlying factor: rising Bitcoin prices. When prices go up, Greed increases, and long trades profit — the relationship may be entirely mediated by price trends.

5. Median PnL Was Zero Across All Sentiments

The median Closed PnL across all sentiment categories was $0.00 (most trades were closed at or near breakeven). Averages are pulled upward by a small number of large winning trades. This means the "average" trader experience may differ substantially from what the mean figures suggest.

6. Survivorship Bias Possible

The 32 traders analyzed are those present in the historical dataset. Traders who lost all capital and stopped trading would not appear in this data. If losing traders are systematically absent, the dataset may present a more favorable picture of overall performance than actually exists across all Hyperliquid users.

7. Fear & Greed Index Limitations

The Fear & Greed Index itself is a proprietary composite score from Alternative.me. Its exact methodology is not fully transparent, and it is a single daily reading applied uniformly to all trades on that day regardless of time of day or individual asset behavior.

Section 12: Key Findings

#	Finding
1	**Extreme Greed produced the highest average PnL per trade.** ***Supporting number:*** *Average PnL = $67.89 during Extreme Greed vs $34.54 during Extreme Fear.* **Why it matters:** Across the sentiment spectrum, the highest-sentiment period consistently produced the best average trade outcome in this dataset.
2	**Win rates were above 76% in all sentiment categories, but this did not guarantee profitability.** ***Supporting number:*** *Win rates: 76.2% (Extreme Fear) to 89.2% (Extreme Greed). Top two profitable traders had win rates of only 33.7% and 36.0%.* **Why it matters:** High win rate strategies are not the only path to profitability. A few large winners can outperform many small winners.
3	**Short positions outperformed long positions significantly during Fear conditions.** ***Supporting number:*** *Close Short average PnL during Fear = $207 vs Close Long = $83.* **Why it matters:** Trade direction appears to matter more than sentiment label alone. During fear periods, being short was historically the better side.
4	**The Fear/Greed score has near-zero correlation with individual trade PnL.** ***Supporting number:*** *Pearson correlation = 0.01 between Fear/Greed Score and Closed PnL.* **Why it matters:** Despite category-level patterns, sentiment score has almost no predictive value at the individual trade level. It is a weak signal.
5	**Losing traders suffered their worst outcomes during Greed, not Fear.** ***Supporting number:*** *Losing trader average PnL during Greed = -$205 per trade.* **Why it matters:** This is counterintuitive and suggests that underperforming traders may have been overly aggressive during bullish periods, not defensive ones.
6	**Fees significantly eroded net PnL for some traders.** ***Supporting number:*** *Trader 0xb899e522... paid $28,643 in fees on $22,489 gross PnL, resulting in net PnL of -$6,155.* **Why it matters:** Fee management is a real concern for active traders. Gross PnL can be misleading without accounting for transaction costs.
7	**The top two traders achieved high PnL with relatively low win rates.** ***Supporting number:*** *0xb123... and 0x0833... had win rates of 33.7% and 36.0% respectively but generated $2.14M and $1.60M total PnL.* **Why it matters:** This pattern is consistent with a high-reward/low-win-rate trading approach where losses are small and winners are large.
8	**Extreme Greed offered the best risk-adjusted return.** ***Supporting number:*** *Sharpe-like ratio = 0.0885 during Extreme Greed vs 0.0304 during Extreme Fear.*

	Why it matters: Not only was average PnL highest during Extreme Greed, but the consistency of returns (relative to variance) was also best.
9	**SOL was the highest-performing coin by average PnL across most sentiment periods.** ***Supporting number:*** *SOL average PnL peaked at ~$285 during Greed conditions and was consistently positive across all sentiment categories.* **Why it matters:** Asset selection may matter as much as sentiment timing. SOL traders in this dataset outperformed traders on other coins.
10	**Trading activity was most concentrated during Fear periods.** ***Supporting number:*** *61,837 trades occurred during Fear — the largest trade count of any sentiment category, despite Fear not being the most common day-type historically.* **Why it matters:** Traders were active during fear periods despite relatively lower average returns, possibly due to perceived buying opportunities.

Section 13: Recommendations

All recommendations below are derived directly from findings in this analysis. None of these are guarantees of future performance. Market conditions in 2024–2025 may not repeat, and all trading involves risk.

1. Consider trade direction as a primary variable in sentiment-based strategies.

Historical data suggests that short positions performed better during Fear periods ($207 average vs $83 for longs) and long positions did better during Greed periods. Traders may consider aligning directional bias with prevailing sentiment rather than simply increasing or decreasing overall trading activity.

2. Monitor fee impact actively — especially for high-frequency strategies.

At least one trader in this dataset lost money net of fees despite positive gross PnL. Traders executing many trades per day should track net PnL (after fees) as the primary performance metric, not gross PnL.

3. Losing traders should exercise extra caution during Greed periods.

Historical data shows that underperforming traders recorded average losses of -$205 per trade during Greed conditions. This pattern indicates that bullish periods may encourage excessive risk-taking among less-disciplined traders. Position sizing discipline during strong uptrends may help limit this.

4. A low win rate does not disqualify a strategy if the reward-to-risk ratio is favorable.

The top two traders both had win rates below 36% but generated over $1.5M each. Traders focused solely on maximizing win rate may be leaving return potential on the table. Reviewing the size distribution of winning vs losing trades is more informative than win rate alone.

5. Avoid over-relying on the Fear & Greed Index as a trade-level signal.

The correlation between sentiment score and individual Closed PnL was 0.01 — effectively zero. The Fear & Greed Index is better used as a broad market context indicator rather than a trigger for individual trade decisions. Category-level patterns exist, but they do not predict the outcome of any specific trade.

6. SOL may warrant closer attention in sentiment-based asset allocation.

Among the top-traded coins in this dataset, SOL consistently showed the highest average PnL across most sentiment periods. Traders who maintain SOL exposure historically benefited from this. However, past performance within a specific dataset does not guarantee future results.

Section 14: Conclusion

Objective

This analysis set out to examine whether Bitcoin market sentiment — as measured by the Fear & Greed Index — has an observable and measurable relationship with trading performance on Hyperliquid. The goal was to generate practical insights from real trade data, not to prove or disprove any particular theory about market psychology.

Major Findings

The most consistent finding across all analysis angles is that sentiment does correlate with average trade outcomes at the category level — but it is a weak predictor of individual trade results. Extreme Greed periods produced the highest average PnL ($67.89), the best win rate (89.2%), and the best risk-adjusted return (0.0885). Extreme Fear produced the lowest win rate (76.2%) and second-lowest average PnL ($34.54).

More importantly, trade direction (Long vs Short) showed a stronger interaction with sentiment than sentiment alone. Short positions averaged $207 during Fear conditions versus $83 for longs — a 2.5x difference. This directional pattern held in the opposite direction during Greed periods.

Statistical Findings

At the individual trade level, the Fear/Greed score had a correlation of 0.01 with Closed PnL — effectively zero. This means that while aggregate patterns exist across sentiment categories, knowing the sentiment score tells you almost nothing about whether any specific trade will be profitable. The distribution of PnL within each sentiment category was broadly similar, with most trades near zero and a small number of large outliers on both sides.

Business Implications

For individual traders, the main takeaway is that trade direction relative to market sentiment may be worth incorporating into a broader strategy framework. For exchange operators and risk managers, the finding that losing traders had their worst outcomes during Greed periods suggests that risk controls during bullish conditions are as important as those during fear periods.

Fees also represent a material cost for active traders. The data shows at least one trader who was net-negative after accounting for fees despite positive gross returns. Fee-aware position sizing is a practical consideration for any high-frequency strategy.

Future Work

- Extending the analysis to a larger trader population would significantly improve the reliability of trader-level findings.
- Including explicit leverage data would allow for more direct analysis of how leverage use changes across sentiment conditions.
- Comparing performance across different market cycles (bull vs bear) would help determine whether the patterns observed in 2024–2025 are cycle-specific or more durable.

- Applying machine learning models with sentiment as a feature — alongside price momentum, volume, and coin type — could quantify how much predictive value sentiment adds when combined with other variables.
- Analyzing holding period (time between trade entry and exit) in relation to sentiment would add a time-dimension that the current analysis does not capture.

End of Report

Bitcoin Market Sentiment vs Trader Performance Analysis | Shubham Sinha | June 2025